Lab Assignment (Final): Product Matching Application using FAISS indexes

Course title : Big Data Analytics
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Table 1: Performance Comparison Between FlatL2 and IVF Indexing Methods

Dataset Size	FlatL2 Build (s)	FlatL2 Search (ms)	IVF Build (s)	IVF Search (ms)	IVF Speedup
4.6K	0.003	0.13	0.167	0.09	1.46x
50K	0.086	2.39	0.679	0.67	3.58x
100K	0.196	2.29	0.823	1.59	1.44x
150K	0.329	3.38	1.001	3.06	1.10x
200K	0.366	3.99	1.162	3.56	1.12x

The following is a comparison of the performance of IVF and FlatL2 indexing techniques, as indicated in Table 1, in regard to different dataset sizes. The two indexing techniques are compared in respect to index construction time, latency, and search speedup. The IVF index tends to consume more time during index construction compared to the FlatL2 index; however, IVF demonstrates faster searches, thus a relatively high search speedup within larger datasets.

Table 2:Performance Analysis of the IVF Index under Different nlist and nprobe Configurations

nlist	nprobe	Build Time (s)	Search Time (ms)	Recall@10	Precision@10
100	10	0.617	0.1	1	1
200	10	0.447	0.05	1	1
100	20	0.275	0.27	1	1
200	20	0.458	0.09	1	1

Table 2 illustrates the effect of varying IVF indexing parameter values, nlist, nprobe, on the cost of index build, efficiency, and effectiveness of search. The build time corresponds to the processing cost involved in generating the IVF index with varying levels of granularity, while the search time corresponds to the average search latency in milliseconds. From the results, the increase in nlist corresponds to increased clustering complexity, adding a slight effect on build time, while the effect of nprobe corresponds to the search path explored per partition, bringing in search latency. However, the search effectiveness remains consistent with all settings, reporting perfect Recall at 10 & Precision at 10.

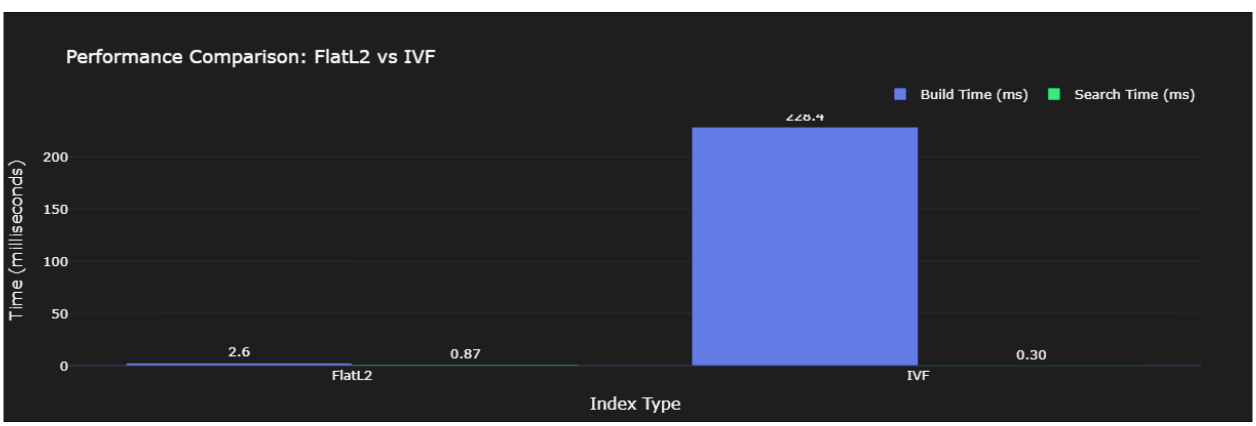

Fig 1:Performance Comparison between FlatL2 and IVF Indexing Methods

This graph shows the comparison between the build time and search time taken by the FlatL2 and IVF indexing methods. FlatL2 has less build time cost and higher search time cost; on the other hand, IVF has high build time cost because of the clustering process and faster search time cost. This graph clearly indicates that there is competition between the cost associated with the indexing method and the search cost.

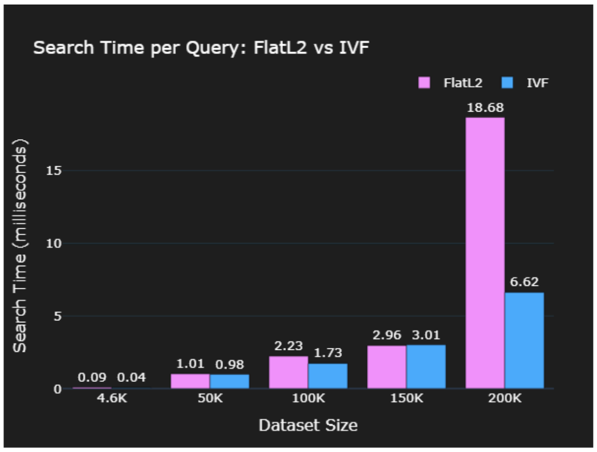

Fig 2: Search Time Comparison between FlatL2 and IVF

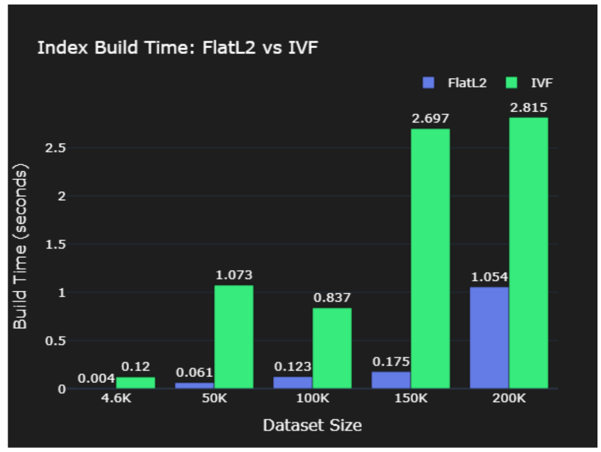

Fig 3: Index Build Time Comparison between FlatL2 and IVF

Figures 2 and 3 present the comparative analysis of FlatL2 and IVF indexing methods with respect to their search time and index build time based on different sizes of the dataset. Figure 2 shows that, for all dataset sizes, the IVF is constantly yielding a lower search time than that of FlatL2. Indeed, as the size increases, at 200K vectors, the performance advantage significantly increases. On the other hand, Figure 3 shows that IVF is consuming more index build time compared to FlatL2, which reflects extra overhead for clustering and index construction. A clear trade-off can be highlighted: IVF is more efficient for large-scale query-intensive applications, while FlatL2 is more suitable when one prefers faster construction of the index and lower preprocessing cost.

IVF build time > FlatL2 build time

IVF Search time < FlatL2 Search time